

A VC-Dimension Generalization Bound via Symmetrization, Sauer–Shelah, and Massart

1 Preliminaries

Notation. Let $\mathcal{X}$ be an arbitrary input space and let $\mathcal{H} \subset \{0, 1\}^{\mathcal{X}}$ be a hypothesis class of binary classifiers. Examples are drawn from $\mathcal{Z} := \mathcal{X} \times \{0, 1\}$ according to an unknown distribution $\mathcal{D}$. We write $S = \{(x_i, y_i)\}_{i=1}^n \sim \mathcal{D}^n$ for an i.i.d. sample. For a measurable $g : \mathcal{Z} \rightarrow [0, 1]$ we use $\mathcal{D}g := \mathbb{E}_{(x,y) \sim \mathcal{D}}[g(x, y)]$ for its population mean and $\widehat{\mathcal{D}}_n g := \frac{1}{n} \sum_{i=1}^n g(x_i, y_i)$ for its empirical mean on S . Throughout, $c, C, C_1, C_2, \dots$ denote universal positive constants whose values may change from line to line and depend on no problem parameter. We adopt the standard measurability convention of empirical-process theory: $\mathcal{H}$ is assumed *image-admissible Suslin* (equivalently, pointwise-separable), so that suprema such as $\sup_{h \in \mathcal{H}} |R(h) - \widehat{R}_n(h)|$ are measurable functions of S and the concentration and symmetrization arguments below apply verbatim.

Definition 1.1 (Population and empirical risk). For $h \in \mathcal{H}$ the population and empirical (training) risks under the 0/1 loss are

$$R(h) := \mathbb{P}_{(x,y) \sim \mathcal{D}}[h(x) \neq y] = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\mathbf{1}\{h(x) \neq y\}], \quad \widehat{R}_n(h) := \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{h(x_i) \neq y_i\}. \quad (1)$$

Definition 1.2 (0/1 loss class). The loss class associated with $\mathcal{H}$ is

$$\mathcal{L} := \{ \ell_h : (x, y) \mapsto \mathbf{1}\{h(x) \neq y\} \mid h \in \mathcal{H} \} \subset \{0, 1\}^{\mathcal{Z}}. \quad (2)$$

With this notation $R(h) = \mathcal{D}\ell_h$ and $\widehat{R}_n(h) = \widehat{\mathcal{D}}_n \ell_h$, so that $\sup_{h \in \mathcal{H}} |R(h) - \widehat{R}_n(h)| = \sup_{\ell \in \mathcal{L}} |\mathcal{D}\ell - \widehat{\mathcal{D}}_n \ell|$.

Definition 1.3 (Empirical Rademacher complexity). Let $\sigma_1, \dots, \sigma_m$ be i.i.d. Rademacher variables, i.e. $\mathbb{P}[\sigma_i = +1] = \mathbb{P}[\sigma_i = -1] = \frac{1}{2}$, independent of the data. For a class $\mathcal{G} \subset \mathbb{R}^{\mathcal{Z}}$ and a sample $T = (z_1, \dots, z_m)$, the empirical and average Rademacher complexities are

$$\widehat{\mathfrak{R}}_T(\mathcal{G}) := \mathbb{E}_{\sigma} \left[\sup_{g \in \mathcal{G}} \frac{1}{m} \sum_{i=1}^m \sigma_i g(z_i) \right], \quad \mathfrak{R}_m(\mathcal{G}) := \mathbb{E}_{T \sim \mathcal{D}^m} [\widehat{\mathfrak{R}}_T(\mathcal{G})]. \quad (3)$$

Definition 1.4 (Growth function). The growth function (shatter coefficient) of $\mathcal{H}$ at size m is

$$\Pi_{\mathcal{H}}(m) := \max_{x_1, \dots, x_m \in \mathcal{X}} |\{ (h(x_1), \dots, h(x_m)) : h \in \mathcal{H} \}|. \quad (4)$$

A set $\{x_1, \dots, x_m\}$ is *shattered* by $\mathcal{H}$ if the maximum above equals 2^m , i.e. all 2^m label patterns are realized. The Vapnik–Chervonenkis dimension $\text{VCdim}(\mathcal{H})$ is the largest m for which some size- m set is shattered.

Assumption 1.5 (I.i.d. sampling). The sample $S = \{(x_i, y_i)\}_{i=1}^n$ consists of n independent draws from a single distribution $\mathcal{D}$ on $\mathcal{Z}$.

Assumption 1.6 (Finite VC dimension, non-degenerate regime). The class $\mathcal{H}$ has $\text{VCdim}(\mathcal{H}) = d < \infty$, and the sample size satisfies $n \geq d \geq 1$.

Remark 1.7. The hypothesis $n \geq d$ in Assumption 1.6 costs nothing. Since both $R(h)$ and $\hat{R}_n(h)$ lie in $[0, 1]$, the deterministic trivial bound $\sup_h |R(h) - \hat{R}_n(h)| \leq 1$ holds for every sample and every n ; in the excluded regime $n < d$ one simply invokes this trivial bound, so no claim about Theorem 5.1 is needed there. The regime $n \geq d$ is exactly where the Sauer–Shelah estimate $\Pi_{\mathcal{H}}(m) \leq (em/d)^d$ is informative and the bound of Theorem 5.1 is the operative one.

Fact 1.8 (The loss class has the same VC dimension). $\Pi_{\mathcal{L}}(m) \leq \Pi_{\mathcal{H}}(m)$ for all m , and hence $\text{VCdim}(\mathcal{L}) \leq \text{VCdim}(\mathcal{H}) = d$.

Proof. Fix $z_1 = (x_1, y_1), \dots, z_m = (x_m, y_m) \in \mathcal{Z}$. The map

$$\psi : (h(x_1), \dots, h(x_m)) \mapsto (\mathbf{1}\{h(x_1) \neq y_1\}, \dots, \mathbf{1}\{h(x_m) \neq y_m\}),$$

is a coordinatewise bijection of $\{0, 1\}^m$: on coordinate i it is the identity when $y_i = 0$ and the bit-flip $b \mapsto 1 - b$ when $y_i = 1$, both bijections of $\{0, 1\}$. A bijection cannot increase the number of distinct images, so the label patterns of $\mathcal{L}$ on $(z_1, \dots, z_m)$ are the image of those of $\mathcal{H}$ on $(x_1, \dots, x_m)$ under this bijection, giving $|\mathcal{L}|_{(z_1, \dots, z_m)}| = |\mathcal{H}|_{(x_1, \dots, x_m)}| \leq \Pi_{\mathcal{H}}(m)$. Taking the maximum over $z_1, \dots, z_m$ yields $\Pi_{\mathcal{L}}(m) \leq \Pi_{\mathcal{H}}(m)$, and a shattered set of $\mathcal{L}$ would force one of $\mathcal{H}$ of the same size, so $\text{VCdim}(\mathcal{L}) \leq d$. This gives the assertion. $\square$

2 Concentration of the uniform deviation

We first reduce the high-probability control of the uniform deviation $\Phi(S) := \sup_{h \in \mathcal{H}} |R(h) - \hat{R}_n(h)|$ to control of its mean, paying only an additive $\sqrt{\log(1/\delta)/(2n)}$. The tool is McDiarmid’s bounded-differences inequality (Theorem 5.1 uses this as its concentration step).

Lemma 2.1 (Bounded-differences concentration of the deviation). *Under Assumption 1.5, let $\Phi(S) := \sup_{h \in \mathcal{H}} |R(h) - \hat{R}_n(h)|$. Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over S ,*

$$\Phi(S) \leq \mathbb{E}_S[\Phi(S)] + \sqrt{\frac{\log(1/\delta)}{2n}}. \quad (5)$$

Proof. We verify that Φ has the bounded-differences property with constants $c_i = 1/n$ and then invoke McDiarmid’s inequality.

Fix a coordinate $i \in [n]$ and two samples $S = (z_1, \dots, z_i, \dots, z_n)$ and $S' = (z_1, \dots, z'_i, \dots, z_n)$ differing only in the i -th example, where $z_j = (x_j, y_j)$. For every fixed $h \in \mathcal{H}$ the population risk $R(h)$ does not depend on S , and the empirical risk changes by at most $1/n$:

$$|\hat{R}_n(h)|_S - \hat{R}_n(h)|_{S'}| = \frac{1}{n} |\mathbf{1}\{h(x_i) \neq y_i\} - \mathbf{1}\{h(x'_i) \neq y'_i\}| \leq \frac{1}{n}, \quad (6)$$

since each indicator lies in $\{0, 1\}$. Writing $a(h) := |R(h) - \hat{R}_n(h)|_S$ and $b(h) := |R(h) - \hat{R}_n(h)|_{S'}$, the reverse triangle inequality and Eq. (6) give $|a(h) - b(h)| \leq 1/n$ for every h , and the elementary inequality $|\sup_h a(h) - \sup_h b(h)| \leq \sup_h |a(h) - b(h)|$ yields

$$|\Phi(S) - \Phi(S')| = \left| \sup_h a(h) - \sup_h b(h) \right| \leq \sup_h |a(h) - b(h)| \leq \frac{1}{n}. \quad (7)$$

Thus Φ satisfies the bounded-differences property with $c_i = 1/n$ for all i , so $\sum_{i=1}^n c_i^2 = n \cdot n^{-2} = 1/n$.

By Assumption 1.5 the examples $z_1, \dots, z_n$ are independent, so McDiarmid's inequality ([1], Theorem 6.2; [4], Corollary 2.21) applies to its upper tail:

$$\mathbb{P}_S \left[\Phi(S) - \mathbb{E}_S \Phi(S) \geq t \right] \leq \exp \left(- \frac{2t^2}{\sum_i c_i^2} \right) = \exp(-2nt^2), \quad t > 0. \quad (8)$$

Setting the right-hand side of Eq. (8) equal to δ gives $t = \sqrt{\log(1/\delta)/(2n)}$; equivalently, with probability at least $1 - \delta$ the complementary event holds, namely

$$\Phi(S) \leq \mathbb{E}_S \Phi(S) + \sqrt{\frac{\log(1/\delta)}{2n}}.$$

This is Eq. (5), as desired. $\square$

3 Symmetrization with a ghost sample

We now bound the expected uniform deviation $\mathbb{E}_S \Phi(S)$ by the Rademacher complexity of the loss class, introducing an independent *ghost sample* S' .

Lemma 3.1 (Ghost-sample symmetrization). *Under Assumption 1.5, with Φ as in Lemma 2.1, $\mathcal{L}$ the loss class of Definition 1.2, and $\bar{\mathcal{L}} := \mathcal{L} \cup (-\mathcal{L})$ its symmetrization,*

$$\mathbb{E}_S [\Phi(S)] = \mathbb{E}_S \left[\sup_{\ell \in \mathcal{L}} |\mathcal{D}\ell - \widehat{\mathcal{D}}_n \ell| \right] \leq 2 \mathfrak{R}_n(\bar{\mathcal{L}}). \quad (9)$$

Proof. Let $S' = (z'_1, \dots, z'_n)$ with $z'_i = (x'_i, y'_i)$ be a ghost sample, i.i.d. $\sim \mathcal{D}$ and independent of S , and let $\sigma_1, \dots, \sigma_n$ be i.i.d. Rademacher signs independent of (S, S') . Write $\ell \in \mathcal{L}$ and recall $\mathcal{D}\ell = \mathbb{E}_{S'} \widehat{\mathcal{D}}'_n \ell$ where $\widehat{\mathcal{D}}'_n \ell = \frac{1}{n} \sum_i \ell(z'_i)$. The proof chains four steps:

$$\begin{aligned} \mathbb{E}_S \left[\sup_{\ell} |\mathcal{D}\ell - \widehat{\mathcal{D}}_n \ell| \right] &\stackrel{(a)}{=} \mathbb{E}_S \left[\sup_{\ell} |\mathbb{E}_{S'} (\widehat{\mathcal{D}}'_n \ell - \widehat{\mathcal{D}}_n \ell)| \right] \\ &\stackrel{(b)}{\leq} \mathbb{E}_{S, S'} \left[\sup_{\ell} |\widehat{\mathcal{D}}'_n \ell - \widehat{\mathcal{D}}_n \ell| \right] = \mathbb{E}_{S, S'} \left[\sup_{\ell} \left| \frac{1}{n} \sum_{i=1}^n (\ell(z'_i) - \ell(z_i)) \right| \right] \\ &\stackrel{(c)}{=} \mathbb{E}_{S, S', \sigma} \left[\sup_{\ell} \left| \frac{1}{n} \sum_{i=1}^n \sigma_i (\ell(z'_i) - \ell(z_i)) \right| \right] \\ &\stackrel{(d)}{\leq} 2 \mathbb{E}_{S, \sigma} \left[\sup_{\ell \in \bar{\mathcal{L}}} \frac{1}{n} \sum_{i=1}^n \sigma_i \ell(z_i) \right] = 2 \mathfrak{R}_n(\bar{\mathcal{L}}), \end{aligned} \quad (10)$$

where $\bar{\mathcal{L}} := \mathcal{L} \cup (-\mathcal{L})$ is the symmetrized loss class. Step (a) rewrites the population mean as the ghost-sample expectation, $\mathcal{D}\ell = \mathbb{E}_{S'} \widehat{\mathcal{D}}'_n \ell$, and uses that $\widehat{\mathcal{D}}_n \ell$ is S' -measurable-free so $\widehat{\mathcal{D}}_n \ell = \mathbb{E}_{S'} \widehat{\mathcal{D}}_n \ell$. Step (b) pulls $\mathbb{E}_{S'}$ outside the absolute value by Jensen's inequality (convexity of $t \mapsto |t|$) and outside the supremum by the monotonicity relation $\sup_{\ell} \mathbb{E}_{S'} \leq \mathbb{E}_{S'} \sup_{\ell}$, so $|\mathbb{E}_{S'}(\cdot)| \leq \mathbb{E}_{S'} \sup_{\ell} |\cdot|$. Step (c) holds because for each i the pair $(\ell(z'_i) - \ell(z_i))$ is symmetric in distribution under the exchange $z_i \leftrightarrow z'_i$, so inserting the independent sign σ_i leaves every joint distribution—and hence the expectation—unchanged. Step (d) first applies the triangle inequality to split the sum and bounds the two resulting suprema by the same quantity (the signs $-\sigma_i$ have the same distribution as σ_i , and S' is an independent copy of S); it then removes the absolute value via the exact identity $\sup_{\ell \in \mathcal{L}} \left| \frac{1}{n} \sum_i \sigma_i \ell(z_i) \right| = \sup_{\ell \in \bar{\mathcal{L}}} \frac{1}{n} \sum_i \sigma_i \ell(z_i)$, which holds because $|t| = \max(t, -t)$ and ranging ℓ over $\bar{\mathcal{L}} = \mathcal{L} \cup (-\mathcal{L})$ realizes both signs; taking $\mathbb{E}_{\sigma}$ yields the (signless) average Rademacher complexity $\mathfrak{R}_n(\bar{\mathcal{L}})$ of Definition 1.3. This proves Eq. (9), which gives the assertion. $\square$

4 Growth function and finite-class Rademacher bound

This section supplies the two combinatorial/analytic ingredients: the Sauer–Shelah bound on the growth function and Massart’s finite-class bound on the Rademacher complexity.

Lemma 4.1 (Sauer–Shelah). *If $\text{VCdim}(\mathcal{H}) = d$, then for every $m \geq 1$,*

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^d \binom{m}{i}, \quad (11)$$

and consequently, for $m \geq d \geq 1$,

$$\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d. \quad (12)$$

Proof. Fix any m points $C = \{x_1, \dots, x_m\} \subseteq \mathcal{X}$ and write $\mathcal{H}_{|C} \subseteq \{0, 1\}^C$ for the set of label patterns realized on C . We prove the stronger *Pajor* claim

$$|\mathcal{H}_{|C}| \leq |\{B \subseteq C : \mathcal{H} \text{ shatters } B\}|, \quad (13)$$

where the empty set is counted as shattered. Since $\text{VCdim}(\mathcal{H}) = d$, no B with $|B| > d$ is shattered, so the right-hand side of Eq. (13) is at most $\sum_{i=0}^d \binom{m}{i}$; Eq. (11) follows by maximizing the left-hand side over C .

We prove Eq. (13) by induction on $m = |C|$. For $m = 0$ both sides equal 1. For the inductive step fix $x \in C$ and set $C' = C \setminus \{x\}$. Apply the *shifting* operator T_x to $\mathcal{H}_{|C}$: for each pattern p with $p(x) = 1$, replace p by the pattern p' agreeing with p off x and with $p'(x) = 0$ whenever $p' \notin \mathcal{H}_{|C}$ (otherwise leave p unchanged). The operator T_x is injective, so it preserves cardinality: $|T_x(\mathcal{H}_{|C})| = |\mathcal{H}_{|C}|$. Partition the shifted family by its restriction to C' : a pattern $q \in (T_x\mathcal{H})_{|C'}$ is counted once if exactly one of $q0, q1$ lies in $T_x\mathcal{H}$, and counted with multiplicity two if both do. The first group has size $|(T_x\mathcal{H})_{|C'}|$, and by the induction hypothesis applied to the family $(T_x\mathcal{H})_{|C'}$ on the $m - 1$ points C' it is bounded by the number of subsets of C' shattered by $T_x\mathcal{H}$, each of which is shattered by $\mathcal{H}$. For the second group, if both $q0$ and $q1$ lie in $T_x\mathcal{H}$ then $T_x\mathcal{H}$ —and hence $\mathcal{H}$ —shatters the set $B \cup \{x\}$ for every $B \subseteq C'$ shattered by the “both” family on C' ; counting these contributes exactly the subsets of C that contain x and are shattered. Adding the two groups, every shattered subset of C is counted once, which is Eq. (13).

For Eq. (12), since $m \geq d$ we have $(d/m)^i \geq (d/m)^d$ for $i \leq d$, so multiplying $\sum_{i=0}^d \binom{m}{i}$ by $(d/m)^d \leq (d/m)^i$ and extending the sum to all $i \geq 0$ gives

$$\left(\frac{d}{m}\right)^d \sum_{i=0}^d \binom{m}{i} \leq \sum_{i=0}^d \binom{m}{i} \left(\frac{d}{m}\right)^i \leq \sum_{i=0}^m \binom{m}{i} \left(\frac{d}{m}\right)^i = \left(1 + \frac{d}{m}\right)^m \leq e^d, \quad (14)$$

using the binomial theorem and $1 + t \leq e^t$. Rearranging Eq. (14) yields $\sum_{i=0}^d \binom{m}{i} \leq (em/d)^d$, which is Eq. (12). This completes the proof. $\square$

Lemma 4.2 (Massart finite-class Rademacher bound). *Under Assumption 1.6, the average Rademacher complexity of the symmetrized loss class $\bar{\mathcal{L}} = \mathcal{L} \cup (-\mathcal{L})$ satisfies*

$$\mathfrak{R}_n(\bar{\mathcal{L}}) \leq \sqrt{\frac{2 \log(2\Pi_{\mathcal{H}}(n))}{n}} \leq \sqrt{\frac{2(d \log(en/d) + \log 2)}{n}}. \quad (15)$$

Proof. Fix a sample $S = (z_1, \dots, z_n)$ and consider the projection $A := \{(\ell(z_1), \dots, \ell(z_n)) : \ell \in \mathcal{L}\} \subseteq \{0, 1\}^n$. The projection of $\bar{\mathcal{L}}$ on S is then $\bar{A} := A \cup (-A)$, with

$$|\bar{A}| \leq 2|A| = 2|\mathcal{L}_S| \leq 2\Pi_{\mathcal{L}}(n) \leq 2\Pi_{\mathcal{H}}(n), \quad (16)$$

by Fact 1.8 and Definition 1.4. Every $a \in A$ lies in $\{0, 1\}^n$, so every $\bar{a} \in \bar{A}$ satisfies $\|\bar{a}\|_2 \leq \sqrt{n}$. Massart's lemma ([2], Theorem 3.7; [3], Lemma 26.8): for a finite $B \subseteq \mathbb{R}^n$ with radius $r = \max_{b \in B} \|b\|_2$ and i.i.d. Rademacher σ ,

$$\mathbb{E}_{\sigma} \left[\max_{b \in B} \sum_{i=1}^n \sigma_i b_i \right] \leq r \sqrt{2 \log |B|}. \quad (17)$$

The empirical Rademacher complexity of $\bar{\mathcal{L}}$ on S therefore satisfies

$$\hat{\mathfrak{R}}_S(\bar{\mathcal{L}}) = \frac{1}{n} \mathbb{E}_{\sigma} \left[\max_{\bar{a} \in \bar{A}} \sum_{i=1}^n \sigma_i \bar{a}_i \right] \stackrel{(a)}{\leq} \frac{r \sqrt{2 \log |\bar{A}|}}{n} \stackrel{(b)}{\leq} \frac{\sqrt{n} \sqrt{2 \log (2\Pi_{\mathcal{H}}(n))}}{n} = \sqrt{\frac{2 \log (2\Pi_{\mathcal{H}}(n))}{n}}, \quad (18)$$

where step (a) is Eq. (17) applied to $B = \bar{A}$ and step (b) uses $r \leq \sqrt{n}$ and Eq. (16). The bound in Eq. (18) holds for every sample S , so taking the expectation $\mathbb{E}_S$ (Definition 1.3) preserves it: $\mathfrak{R}_n(\bar{\mathcal{L}}) \leq \sqrt{2 \log (2\Pi_{\mathcal{H}}(n))/n}$. Finally Lemma 4.1 with $m = n \geq d$ gives $\log \Pi_{\mathcal{H}}(n) \leq d \log(en/d)$, so $\log(2\Pi_{\mathcal{H}}(n)) \leq d \log(en/d) + \log 2$, yielding the second inequality in Eq. (15). This gives the assertion. $\square$

5 The VC generalization bound

We assemble Lemmas 2.1, 3.1, 4.1 and 4.2 into the uniform generalization bound. The proof is a short chain of cross-references; all the work was done in the preceding lemmas.

Theorem 5.1 (Uniform VC generalization bound). *Suppose Assumptions 1.5 and 1.6 hold, so that $\mathcal{H} \subset \{0, 1\}^{\mathcal{X}}$ has $\text{VCdim}(\mathcal{H}) = d$ and $n \geq d \geq 1$. Then there is a universal constant $C > 0$ such that for every distribution $\mathcal{D}$ and every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the i.i.d. sample $S \sim \mathcal{D}^n$,*

$$\sup_{h \in \mathcal{H}} |R(h) - \hat{R}_n(h)| \leq C \sqrt{\frac{d \log(en/d) + \log(1/\delta)}{n}} = C \sqrt{\frac{d \log(n/d) + d + \log(1/\delta)}{n}}. \quad (19)$$

In particular the leading complexity term is the VC rate $d \log(n/d)$ of the prompt, the additive d being the explicit form of the constant $\log e$ inside $\log(en/d)$.

Proof. Write $\Phi(S) = \sup_{h \in \mathcal{H}} |R(h) - \hat{R}_n(h)|$ as in Lemma 2.1.

Failure budget (union bound). The high-probability content of Eq. (19) comes from the single concentration event of Lemma 2.1. Define the favorable event

$$\mathcal{E} := \left\{ \Phi(S) \leq \mathbb{E}_S \Phi(S) + \sqrt{\frac{\log(1/\delta)}{2n}} \right\}. \quad (20)$$

By Lemma 2.1, $\mathbb{P}[\mathcal{E}] \geq 1 - \delta$; equivalently the complementary (failure) event has probability at most δ . As this is the only stochastic step—Lemmas 3.1, 4.1 and 4.2 are deterministic statements about the fixed class $\mathcal{H}$ —a union bound over the single event $\mathcal{E}$ already accounts for the entire failure budget δ . We condition on $\mathcal{E}$ for the remainder of the proof.

Deterministic bound on $\mathcal{E}$. On $\mathcal{E}$ we bound the mean deviation by composing the symmetrization and finite-class steps:

$$\begin{aligned}
\Phi(S) &\stackrel{(a)}{\leq} \mathbb{E}_S \Phi(S) + \sqrt{\frac{\log(1/\delta)}{2n}} \\
&\stackrel{(b)}{\leq} 2\mathfrak{R}_n(\bar{\mathcal{L}}) + \sqrt{\frac{\log(1/\delta)}{2n}} \\
&\stackrel{(c)}{\leq} 2\sqrt{\frac{2(d\log(en/d) + \log 2)}{n}} + \sqrt{\frac{\log(1/\delta)}{2n}} \\
&\stackrel{(d)}{\leq} \frac{4\sqrt{d\log(en/d)} + \sqrt{\log(1/\delta)}}{\sqrt{n}} \stackrel{(e)}{\leq} \frac{C_0\sqrt{d\log(en/d) + \log(1/\delta)}}{\sqrt{n}}, \tag{21}
\end{aligned}$$

where (a) is the definition Eq. (20) of $\mathcal{E}$; (b) is the symmetrization bound $\mathbb{E}_S \Phi(S) \leq 2\mathfrak{R}_n(\bar{\mathcal{L}})$ of Lemma 3.1; (c) is the Massart/Sauer–Shelah bound $\mathfrak{R}_n(\bar{\mathcal{L}}) \leq \sqrt{2(d\log(en/d) + \log 2)/n}$ of Lemma 4.2; (d) collects the common factor $1/\sqrt{n}$ and uses $d\log(en/d) + \log 2 \leq 2d\log(en/d)$ (valid since $d\log(en/d) \geq d \geq 1 > \log 2$), so $2\sqrt{2}\sqrt{d\log(en/d) + \log 2} \leq 4\sqrt{d\log(en/d)}$; and (e) uses $\alpha\sqrt{u} + \beta\sqrt{v} \leq \sqrt{\alpha^2 + \beta^2}\sqrt{u+v}$ (Cauchy–Schwarz in $\mathbb{R}^2$) with $\alpha = 4$, $\beta = 1$, so that $C_0 = \sqrt{4^2 + 1^2} = \sqrt{17}$.

Taking $C := C_0 = \sqrt{17}$ and recalling $\log(en/d) = \log(n/d) + 1$, the bound Eq. (21) reads, on $\mathcal{E}$,

$$\Phi(S) \leq C\sqrt{\frac{d\log(en/d) + \log(1/\delta)}{n}} = C\sqrt{\frac{d\log(n/d) + d + \log(1/\delta)}{n}}. \tag{22}$$

Since $\mathbb{P}[\mathcal{E}] \geq 1 - \delta$ by the failure-budget paragraph above, Eq. (22) holds with probability at least $1 - \delta$, which is exactly Eq. (19). This completes the proof. $\square$

Remark 5.2 (On the rate and the constant). The bound Eq. (19) is the classical VC rate $\tilde{O}(\sqrt{d/n})$: the $d\log(n/d)$ term is the metric entropy of $\mathcal{H}$ at scale n (via Sauer–Shelah), and the $\log(1/\delta)$ term is the price of high-probability control (via McDiarmid). We did not optimize the universal constant $C = \sqrt{17}$; a sharper chaining (Dudley) argument removes the $\sqrt{\log(n/d)}$ factor and yields the optimal $\sqrt{d/n}$ rate, but that refinement is outside the stated symmetrization/Sauer–Shelah/Massart pipeline.

References

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